

MECHANISTIC INTERPRETABILITY: RESULTS AND DESIGN CHANGES

FinPersona-Bench | what the M1 to M4 results show, what we expected, and what to change | 23 Aug 2026

PART 1. WHAT THE RESULTS SHOW, AND WHAT WE EXPECTED

In one sentence. The internals say that re-injecting the mandate works because the copy sits at the end of the prompt where the model looks most, and that the persona is a single steerable direction. They do not say that the mandate fades over time. We expected fading; we found recency.

M1. Attention to the mandate

What we expected. Attention to the mandate would fall as the rollout goes on. Memory would stop the fall. This was supposed to be the main figure for Mandate Salience Decay.

What we found. Attention to the mandate is flat for all 200 days in every model. Memory does not raise attention to the original mandate at all. It adds a second copy at the end of the prompt, and that copy gets about as much attention as the original, so the total roughly doubles (1.5 to 1.8 times).

What it means. The flat line is not a discovery. Each day is a fresh prompt with the same structure, so nothing can build up and nothing can fade. The doubling is a position effect: the end of the prompt is where the model looks. Two details matter more than the headline. First, memory takes attention away from the market data block (about one third less). That is a real reason why memory can hurt in the bubble: the agent reads the market less. Second, the format instructions get three to five times more attention than the persona in every arm. The persona is a small part of what the model attends to.

M1 scaling (Qwen 1.5B to 14B)

What we expected. We had no clear expectation. The plan did not include this test.

What we found. The memory-to-static ratio stays at 1.4 to 1.8 times across all sizes. Absolute attention to the mandate does not grow with size either.

What it means. The recency effect is not a quirk of one model. Bigger models do not attend more to the persona.

M1 growing context (the only real decay test)

What we expected. With a prompt that actually grows, attention to the mandate at the front would fall.

What we found. For one model, one persona, one scenario and 40 days (about 2.5k tokens), the front mandate gets very little attention (under one percent) and does not fall. The re-injected copy at the end gets 3.7 times more than the front copy.

What it means. This is again recency, not decay. But the test is too short to say decay does not exist. Real decay in the literature shows up over many turns and much longer contexts. Llama-3.1 can hold 128k tokens; the 4k limit in this test was self-imposed. The decay question is still open and needs the stateful agent with the full 200-day context.

M2. A linear direction for the mandate

What we expected. One direction in the activations that tracks adherence, which we could read every step as a drift signal and later use to trigger re-grounding.

What we found. Two different directions were built. The mandate-presence direction (persona prompt versus neutral prompt) is clean and is the one used for steering. The adherence direction (adherent steps versus violating steps) separates the two almost perfectly at layer 0, which means it is reading the cash number printed in the prompt, not an internal state. Its transfer to the crash scenario is weak (0.55 to 0.78).

What it means. We have a direction for how strongly the persona is present, not a direction for whether the agent is drifting. There is also an unexplained result in the projection figure: the static arm projects more strongly on the mandate-presence direction than the memory arm, in all three scenarios. If re-injection worked through this direction, memory should project more, not less. The most likely cause is that the direction was fitted on static-style prompts only. Until this is resolved, the claim that the text reminder and the internal direction act through the same channel is not supported.

M3. Steering the direction (the key result)

What we expected. Adding the direction would improve adherence, removing it would worsen adherence, with the prompt unchanged, and the regime sign-flip from the benchmark would appear.

What we found. All three happen. Adding the direction lowers MAS step by step, removing it raises MAS, and the sign-flip appears in all three models: steering lowers Rationality in the bubble and keeps or raises it in the crash.

What it means. The causal handle is real, but what it controls is simpler than the headline. For ENTJ, MAS stops at exactly 0.20 once alpha reaches 0.10. Under the old point target of 0.2 cash, MAS 0.20 means cash is zero: the steered agent goes all-in and stays there. At negative alpha, MAS is 0.80, which means cash is one: the agent never buys. So the direction is a risk-on / risk-off knob between two extremes, not a calibrated adherence control. For ISFJ and INTJ the curves are almost flat, so this is mainly an ENTJ result. The sign-flip follows from that knob and the old Rationality rule: in the old crash the price is always below value, so buying is always scored as rational even while money is lost; in the bubble the price is above value, so buying is scored as irrational. Push the agent to buy everything and Rationality goes up in the crash and down in the bubble by construction. The activation intervention reproduces the benchmark pattern, but the pattern comes largely from the metric, not from a regime-aware mechanism. This is the same problem that led us to replace Rationality with mandate-conditional regret.

M4. Attribution for the mini-model crash reversal

What we expected. Logit attribution would show where memory rescues crash behaviour in small models.

What we found. The reversal (memory improves crash Rationality) exists only below about 4B parameters (Gemma-2-2B +100, Gemma-3-4B +12). At 7 to 9B memory makes Rationality worse in every regime. In the 2B model, the static agent holds all cash and never trades; memory makes it buy, and the old rule scores buying in a crash as rational. The change is written at layer 19.

What it means. The gate worked and produced a useful negative: the mini-model reversal is a small-model effect, not a general one. The word 'rescue' should go: the small model is being nudged to buy into a falling market. The attribution itself (layer 19, static favours HOLD, memory neutralises it) is a correct description of that nudge.

Circuit: read early, integrate mid, decide late

What we expected. Not planned.

What we found. Layer-0 heads read the mandate, mid-stack heads (around layers 7 and 12 to 14) are boosted by memory, and restoring the direction at layer 7 alone recovers most of the persona action. The restore-at-layer curve is noisy and based on 20 days.

What it means. A plausible sketch, worth keeping as descriptive material, not yet a result.

Summary

Experiment	We expected	We found	Verdict
M1 attention	Attention to the mandate decays; memory stops the decay	No decay; memory adds a second copy at the end that doubles total attention and takes attention from the market block	Opposite to the premise, for a structural reason (stateless agent)
M1 scaling	No expectation	Ratio stable 1.4 to 1.8x from 1.5B to 14B	Solid
M1 growing context	Front mandate attention falls as context grows	Flat over 40 days / 2.5k tokens; recent copy gets 3.7x	Recency shown; decay untested at real lengths
M2 directions	One adherence direction usable as a drift signal	Presence direction is clean; adherence direction reads the prompt; static projects higher than	Partly opposite; same-channel claim unsupported

Experiment	We expected	We found	Verdict
		memory (unexplained)	
M3 steering	Direction controls adherence; sign-flip reproduced through the same channel	Direction is a risk-on/off knob; sign-flip reproduced but largely via the old Rationality rule; ENTJ-driven	Confirmed on the surface, weaker underneath
M4 attribution	Explains the mini-model crash reversal	Reversal only below ~4B; 'rescue' is buying into a crash; located at layer 19	Confirmed, with a caveat that changes the claim
Circuit	Not planned	Read at layer 0, integrate around layer 7, decide late; noisy	Descriptive only

What this means for the paper. The internals support the thesis that re-grounding is a directive amplifier placed where the model looks, and that persona intensity is a steerable direction. They do not support salience decay on the current architecture. The mechanistic story and the behavioural story now agree, but neither is the decay story the original paper told.

PART 2. CHANGES NEEDED TO FIT THE NEW SYNTHETIC ENVIRONMENT

Everything in the report was run on the old environment, the old prompts, the old action space and the old metrics. These are the changes needed before any mechanistic result can be re-run or reported.

	Area	In the report	Change needed	Why
1	Prompt replay	Replays the old 10-field prompt and verifies by byte comparison	Rebuild replay on the new renderer (19 market fields plus cash share); verify by reproducing the logged prompt hash and config hash	Every span offset changes; the hash is the new proof that the replayed prompt is the real one
2	Token spans	Mandate, market, placebo (JSON format text), memory_refresh, decision	Regenerate all spans; add spans for the new arms (declarative placebo, directive placebo, wrapper-only, swapped mandate) and for the analyst fair-value field	The arms are new text blocks; the analyst field is the one displayed value signal
3	Action read-out	BUY / SELL / HOLD label; verified 20 of 20	Read the numeric target cash share; redo the read-out check for number parsing; fix the query position (first digit token) and use it everywhere	The action is a number now, not one of three words
4	M4 logit target	SELL minus HOLD logit	Define a numeric read-out (for example high-cash versus low-cash digit token sets, or target above versus below current share) before any attribution	The SELL and HOLD logits no longer exist as the decision
5	M4 gate	Reversal checked with old start (100 percent cash) and old action space	Re-check which models show the crash reversal under start-at-band-centre and the target-share action	The old 'frozen cash, HOLD paralysis' story depends on the old start and the SELL no-op
6	Adherence labels (M2)	Deviation from point targets 0.2 / 0.5 / 1.0; adherent at or below 0.15, violating at or above 0.35	Redefine on band deviation with centres 0.10 / 0.50 / 0.80; choose thresholds again; rebalance	Targets and metric changed
7	Behavioural scores (M3, M4)	Old MAS, old Rationality, gross return	Re-score with band-MAS, mandate-conditional regret on resolvable steps, drawdown, return net of costs; report against the trivial-policy floor and the oracle ceiling	Old Rationality produced the sign-flip by construction; flat now has resolvable steps
8	Time alignment	Curves plotted by calendar day; scenarios have fixed phase timing	Align by the generator's phase labels; report per phase; stratify bull trap by topped / not topped and crash by severity	Phase onsets are randomised per seed, so day number no longer means the same thing
9	Scenarios	Flat, bubble, crash	Add sustained bull to M3	It is a falsification test: steering should move adherence but not regret when there is no mispricing
10	Start design	All runs start at 100 percent cash	Use start-at-band-centre for the main runs and the common-start (0.5) slice for training directions	Persona and portfolio state are correlated at band-centre start; the common-start slice breaks that

	Area	In the report	Change needed	Why
11	Portfolio state in prompt	Cash and holdings shown	Cash share is now printed explicitly; treat it as a known confound for any adherence direction	The M2 adherence direction already reads this number
12	Arms for M1 and M2	Static versus memory only	Run M1 on every arm; use swapped-mandate pairs (same wrapper, opposite content) as the contrast for M2	Gives the content-versus-force test inside the model and a cleaner direction
13	Stateful agent	Growing-context probe limited to 4k tokens, measurement only, 40 days	Replay the real stateful agent's full 200-day context (observations, decisions, rationales); measure attention, projection and behaviour together; exclude restatement-probe turns	This is the only place decay can be tested
14	Decoding	Greedy generation in steered rollouts	Match the harness (temperature 0.2, at least 3 decodes) or log greedy as an explicit factor	Results must be comparable with the behavioural grid
15	Provenance	No hashes in mechanistic outputs	Stamp environment version, config hash and prompt hash into every mechanistic output row; use the same seeds as the grid	Same standard as the behavioural runs

PART 3. CHANGES TO THE EXPERIMENTS THEMSELVES: CONTROLS, ROBUSTNESS, VARIABILITY

These apply regardless of the environment. They are ordered by how much they change what we can claim.

	Change	What to do	Why it matters	Priority
1	Mediation test for 'same channel'	Ablate the mandate-presence direction inside the memory arm and check whether the memory benefit disappears; check whether static plus the direction reproduces the memory arm's allocation path	M3 shows the direction can move behaviour; it does not show that re-injection works through it. This is the test that would	High
2	Explain the projection reversal	Refit the direction on both prompt forms, or on the swapped-mandate contrast; re-plot static versus memory projections	Static currently projects higher than memory on the mandate direction, which contradicts the mechanism story	High
3	Recency-position placebo in M1	Place a length-matched neutral block at the re-injection position; measure its attention and its effect on behaviour	If any recent block gets 2 to 4 times the attention, the doubling is positional. The content effect is then the difference between recent placebo and recent mandate	High
4	Report allocation, not only MAS	Plot cash share against alpha directly; mark where the agent saturates at all-in or all-cash	The MAS plateau at 0.20 hides that the agent went all-in; readers need to see the saturation	High
5	Characterise the direction	Compare the mandate-presence direction with an aggressive-versus-conservative direction, a generic trade-versus-hold direction, and persona-specific directions for ISFJ and INTJ; report cosine similarities	We need to know whether we are steering 'persona' or just 'invest more'	High
6	Random and orthogonal steering controls	Steer a random direction and an orthogonal direction at the same norm-relative alpha; report behaviour and output validity (parse rate, rationale coherence)	Without these, any push of that size might move behaviour; alpha at or above 0.2 already breaks generation	High
7	Adherence direction that can drive a controller	Label adherence by the next action's deviation, not the printed cash share; or project out the part predictable from the cash tokens; train on the common-start slice; test on held-out scenario and persona; check whether it moves over time on the stateful arm	The current adherence direction reads the input; a drift signal must reflect internal state	High
8	Seeds, models and uncertainty	At least 3 seeds per model and persona for M3 (Gemma and Qwen are currently 1 seed, ENTJ only); confidence intervals on every ratio and curve; anchor models chosen from those that pass the $t = 0$ gate	Most of the sign-flip evidence outside Llama is $n = 1$	High
9	Sustained-bull	Run M3 on the sustained-bull scenario	Steering should change adherence but	High

	Change	What to do	Why it matters	Priority
	falsification		not regret; if it changes regret, the direction is doing more than persona intensity	
10	Layer sweep for steering	Sweep the injection layer for M3 as was done for restore-at-layer; report effect size per layer	The mid-layer band was chosen by hand	Medium
11	Per-layer and sink-corrected attention	Report attention per layer at the decision position, not pooled; normalise by one minus the sink share	Pooling hides the read-early / integrate-mid structure the circuit section relies on; Llama puts half of attention on the first token	Medium
12	Value-weighted attention	Weight attention by the value-vector norm, or use attention rollout, as a second measure	Raw attention pattern says where the model looks, not what it reads	Medium
13	Stronger M4 method	Use path patching or activation patching of the mandate span into the static run; use many more crash days than 12; attribute both directions of the decision	Layer-wise attribution differences do not establish a path from the mandate to the decision	Medium
14	Persona coverage	Run the full M3 grid for ISFJ and INTJ, not only ENTJ; fit persona-specific directions	The headline generalises from ENTJ; the other personas are nearly flat	Medium
15	Mandate wording variants	Repeat M1 and M2 with the paraphrased and abbreviated mandate texts	Shows whether the direction and the attention effect depend on exact wording	Medium
16	Steering operating range	Define the valid alpha range per model from output validity, and report only inside it	Alpha at 0.2 and below minus 0.1 collapse the model; the range must be set before the results are read	Medium
17	Restore-at-layer robustness	Repeat the restore sweep over more days and seeds; report intervals	The current curve is one spike on 20 days	Low
18	Attention-to-market as a result	Report the drop in attention to the market block under memory as its own finding, per regime	It is a candidate mechanism for the bubble penalty and is already in the data	Low

What stays as it is. The replay verification, the placebo / uniform / first-token-sink normalisation, the scaling study, the ablate-and-restore protocol and the per-head analysis are well built and carry over unchanged. The changes above are about what the steered direction means, what the behavioural readouts are scored with, and the two missing tests (recency placebo and mediation) that separate 'amplifier' from 'same mechanism'.